Globally consistent mapping for indoor and outdoor environments using Hybrid LiDAR SLAM

Zewdie Habtie Sisay

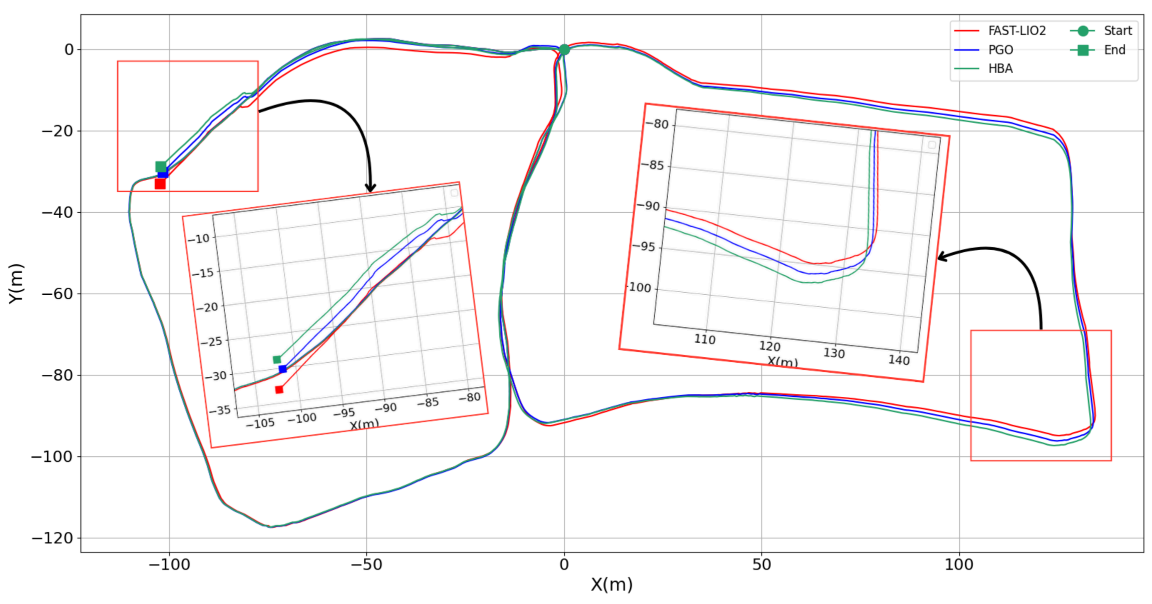
Supervisors: Prof. Zoltan Istenes, Prof. Mohammad Aldibaja

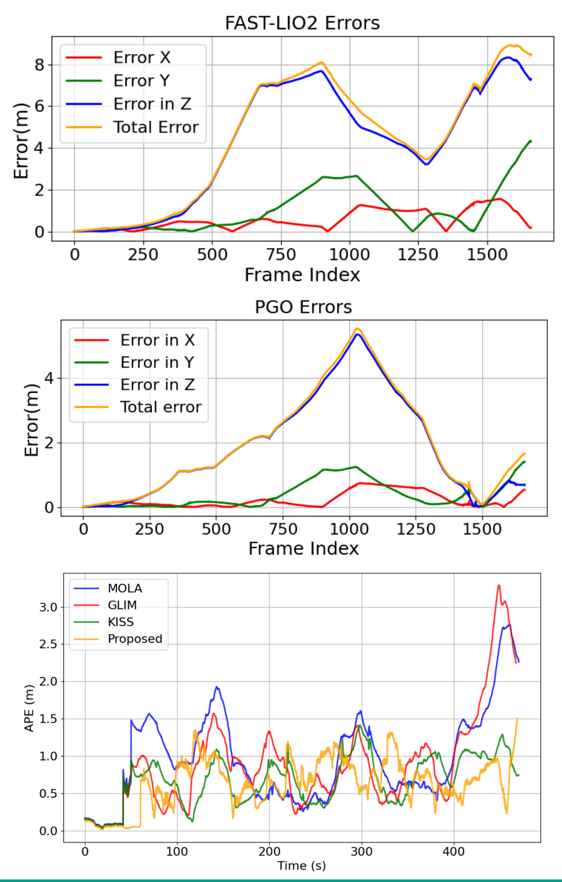
Introduction

An accurate map is an important prerequisite for autonomous robots to navigate, plan paths, localize, avoid obstacles and execute tasks efficiently. Simultaneous Localization and Mapping (SLAM) algorithms using LiDAR point clouds are the most preferred mapping method due to the accuracy of the 3D range measurements of LiDAR sensors.

Problem statement: Due to an inherent noise in sensor measurements, robot motion, and dynamics of the environment, inconsistencies of the map generated with LiDAR SLAM are inevitable. An inconsistent map of the environment causes accumulation of localization errors, failure in path planning, increased risk of collision, and mission failure. Since SLAM is not closed-form solution, generating 100% accurate map of the environment remains a challenge.

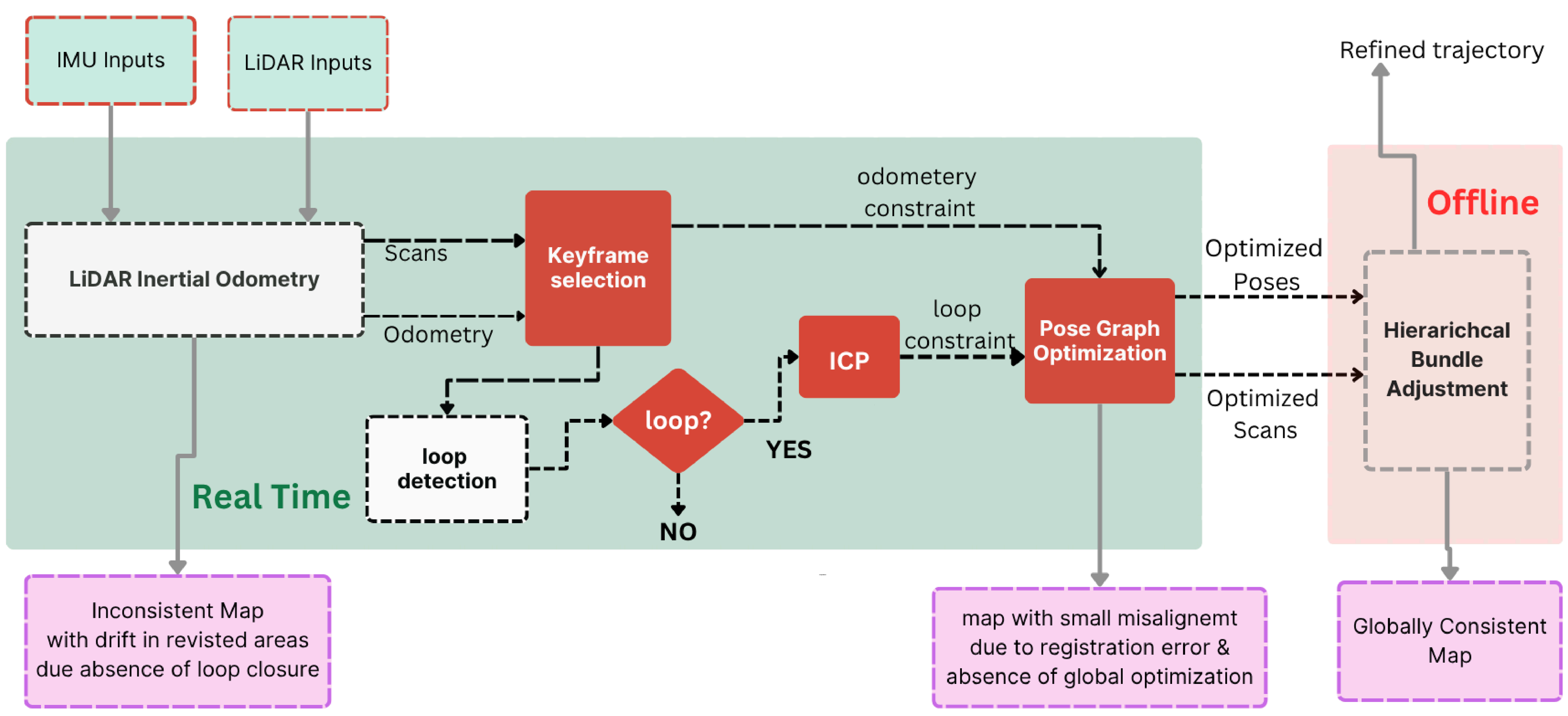
Trajectory Errors





FAST-LIO2 PGO HBA Large scale map

Methodology



FAST-LIO2

Prediction

$$\hat{\mathbf{x}}_{i+1} = \hat{\mathbf{x}}_i \boxplus \left(\Delta t \mathbf{f}(\hat{\mathbf{x}}_i, \mathbf{u}_i, \mathbf{0})\right); \quad \hat{\mathbf{x}}_0 = \bar{\mathbf{x}}_{k-1},$$

$$\hat{\mathbf{P}}_{i+1} = \mathbf{F}_{\bar{\mathbf{x}}_i} \hat{\mathbf{P}}_i \mathbf{F}_{\bar{\mathbf{x}}_i}^T + \mathbf{F}_{\mathbf{w}_i} \mathbf{Q}_i \mathbf{F}_{\mathbf{w}_i}^T; \quad \hat{\mathbf{P}}_0 = \bar{\mathbf{P}}_{k-1}$$

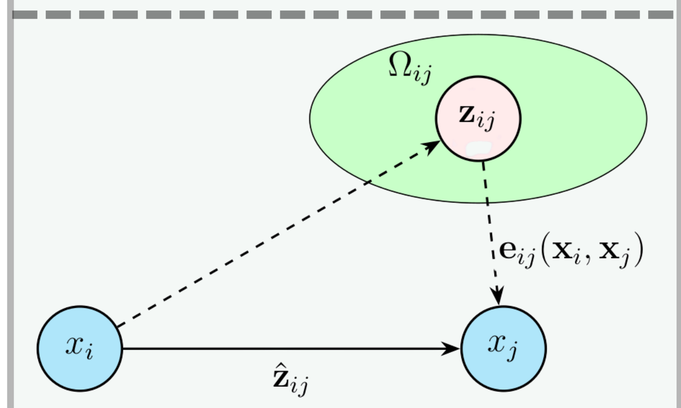
Iterated Update

$$\mathbf{z}_j^\kappa = \mathbf{u}_j^T \left({}^G\hat{\mathbf{T}}_{I_k}^\kappa \, {}^I\hat{\mathbf{T}}_{L_k}^\kappa \, {}^L\mathbf{p}_j - {}^G\mathbf{q}_j\right),$$

$$\mathbf{K} = \left(\mathbf{H}^T\mathbf{R}^{-1}\mathbf{H} + \mathbf{P}^{-1}\right)^{-1}\mathbf{H}^T\mathbf{R}^{-1},$$

$$\hat{\mathbf{x}}_k^{\kappa+1} = \hat{\mathbf{x}}_k^\kappa \boxplus \left(-\mathbf{K}\mathbf{z}_k^\kappa - (\mathbf{I} - \mathbf{K}\mathbf{H})(\mathbf{J}^\kappa)^{-1}(\hat{\mathbf{x}}_k^\kappa \boxminus \hat{\mathbf{x}}_k)\right)$$

PGO

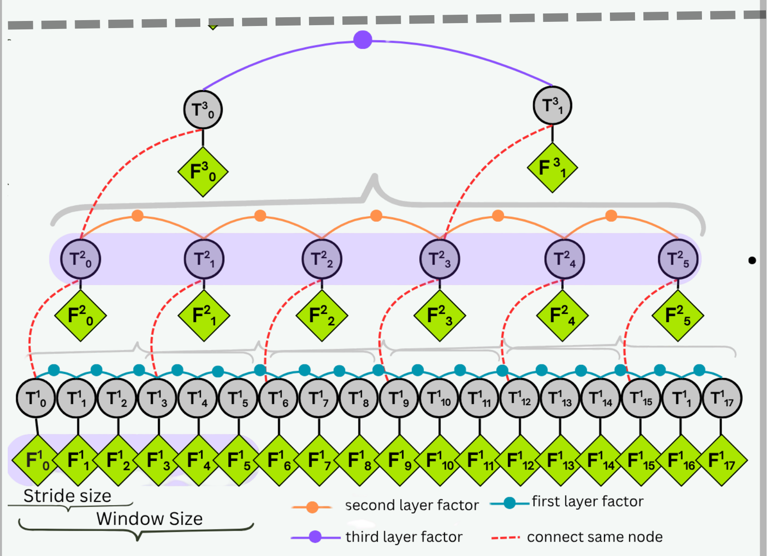


$$\mathbf{e}_{ij}(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{z}_{ij} - \hat{\mathbf{z}}_{ij}(\mathbf{x}_i, \mathbf{x}_j)$$

$$\mathbf{F}(\mathbf{x}) = \sum_{\langle i,j\rangle \in \mathcal{C}} \underbrace{\mathbf{e}_{ij}^T \boldsymbol{\Omega}_{ij} \mathbf{e}_{ij}}_{\mathbf{F}_{ij}}$$

$$\mathbf{x}^* = \arg\min_{\mathbf{x}} \mathbf{F}(\mathbf{x})$$

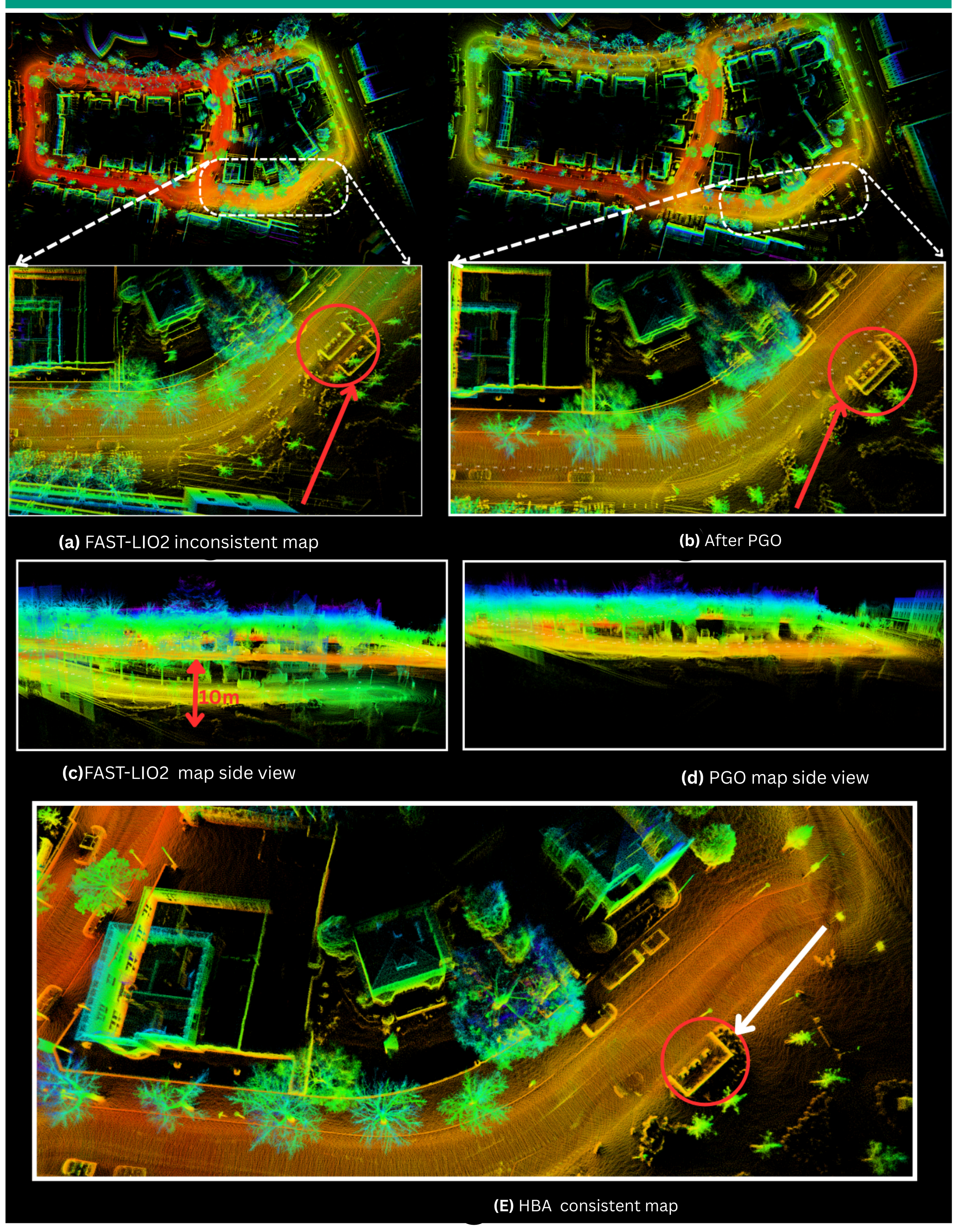
HBA



$$\mathbf{T}_j^{i+1} = \mathbf{T}_{s,j}^{i*} = \prod_{k=1}^{j} \mathbf{T}_{s,k-s,s,k}^{i*}$$

$$\mathbf{F}_j^{i+1} \triangleq \bigcup_{k=0}^{w-1} \left(\mathbf{T}_{s,j,s,j+k}^{i*} \cdot \mathbf{F}_{s,j+k}^{i}\right)$$

Map quality



(a) FAST-LIO2 inconsistent map (b) After PGO (c) FAST-LIO2 map side view (d) PGO map side view (E) HBA consistent map

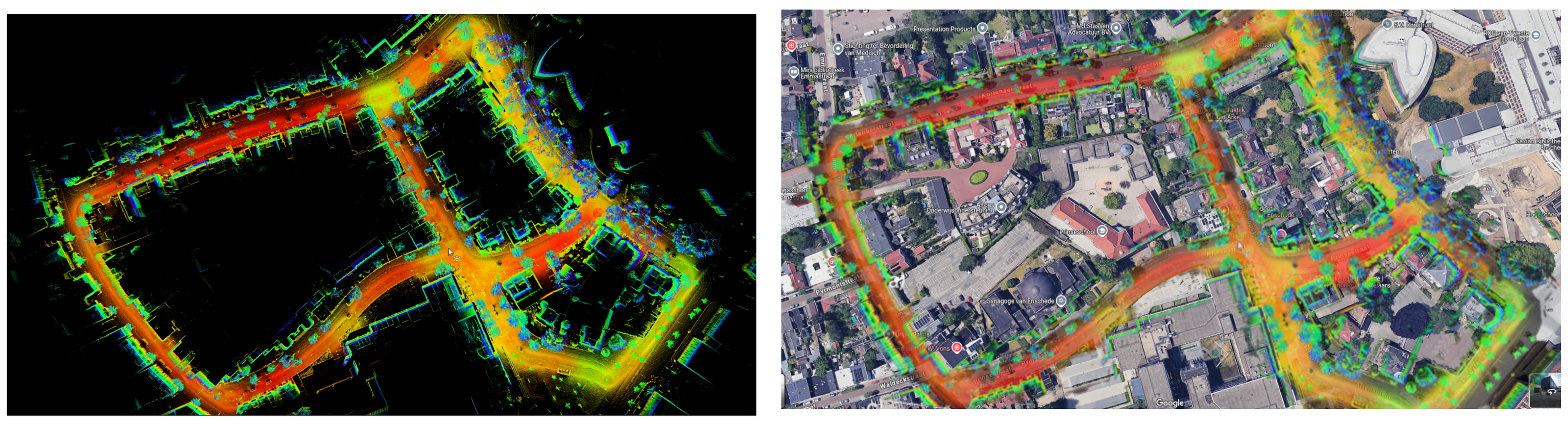
Mean Map Entropy(MME) of the three methods across different datasets

Method	S01	S02	S03	S11	K03	$\mathcal{K}$00
FAST-LIO2	-2.63, 0.11	-1.9, 0.19	-2.46, 0.19	**-3.34**, 0.10	-1.85, 0.07	-2.40, 0.04
PGO	<u>-2.66</u>, 0.14	**-2.50**, 0.12	<u>-2.60</u>, 0.30	<u>-3.21</u>, 0.20	<u>-1.91</u>, 0.03	<u>-2.47</u>, 0.08
HBA	**-2.71**, 0.05	<u>-2.48</u>, 0.084	**-2.85**, 0.13	-3.20, 0.091	**-2.39**, 0.05	**-2.97**, 0.026

S stands for Saxion, K stands for KAIST, $\mathcal{K}$ stands for KITTI

Contributions

FAST-LIO2 ensures local consistency but **accumulates drift** over long trajectories, leading to global map errors. Pose Graph Optimization (**PGO**) with loop closure reduces this drift but still leaves small inconsistencies due to registration errors, insufficient (false) loop closures, or a wrong initial guess. To correct these, Hierarchical Bundle Adjustment (**HBA**) is applied. Since HBA cannot converge on large inconsistencies, PGO acts as an interface between FAST-LIO2 and HBA, minimizing drift to within HBA's convergence range. Our main contribution lies on bridging FAST-LIO2 and HBA incorporating loop closure with PGO.



Conclusion

The proposed SLAM system significantly improved trajectory accuracy and map consistency across datasets. PGO reduced real-time drift, while HBA enhanced global alignment and consistency. The hybrid approach performed well in indoor and large-scale outdoor environments with frequent loop closures, resulting in a robust and consistent mapping pipeline.